Subspace Alignment for CLIP-based Continual Learning via Canonical Correlation Analysis

Addressing Asymmetric Drift in Vision-Language Continual Learning

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Agenda

1. **Problem Definition** — Asymmetric Drift & Modality Misalignment in CLIP-CL
2. **Theoretical Foundations** — Canonical Correlation Analysis (CCA) Principles
3. **Proposed Solution (CCA-CL)** — Statistical Memory, Subspace Projection & RFP Extension
4. **Experimental Results & Ablation** — SOTA Benchmark Comparison, Efficiency, and Sensitivity Analysis

1. Problem Definition: Continual Learning with CLIP

Exemplar-Free Class-Incremental Setting

- **Exemplar-Free:** Past raw samples are completely inaccessible (privacy/memory bounds).
- **Class-Incremental:** Task classes arrive sequentially in disjoint sets ($C_i \cap C_j = \emptyset$).
- **Task-ID Unknown:** Inference evaluates across all seen classes without task labels.

CLIP Zero-Shot Classification

- **Visual Feature:** $z_x = F_v(x)$ via Visual Encoder.
- **Text Prototype:** $z_y^{(c)} = F_t(\text{"a photo of [CLS]"})$ via Text Encoder.
- **Cosine Similarity Prediction:**

$$\hat{y} = \arg\max_c \mathrm{sim}(z_x, z_y^{(c)}) = \frac{z_x^\top z_y^{(c)}}{||z_x||_2||z_y^{(c)}||_2}$$

Core Challenge: Adapting pre-trained Vision-Language Models to sequential streams without destroying cross-modal alignment or retaining past data samples.

1. Problem Definition: The Asymmetric Drift (AD) Dilemma

Unimodal adaptation creates a fundamental imbalance between visual and text modalities.

Visual Encoder (Dynamic)

- **High Input Variability:** Visual data distribution shifts significantly across task streams.
- **Drastic Parameter Updates:** Visual feature representations change rapidly during fine-tuning.

Text Encoder (Static)

- **Low Prompt Variance:** Text prompts use fixed templates ("a photo of [CLS]").
- **Minimal Updates:** Text feature representations remain essentially stationary.

Asymmetric Drift (Result)

- **Widening Modality Gap:** Visual features drift while text features stay fixed.
- **Representation Collapse:** Cross-modal feature distances expand, leading to classification failure.

1. Problem Definition: Asymmetric Drift Concept & Formulation

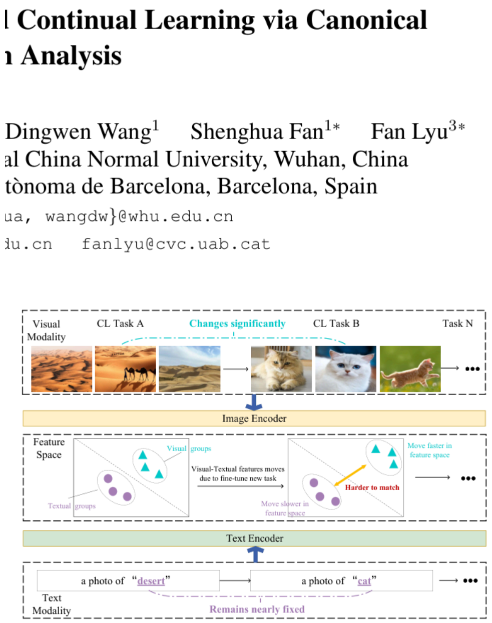
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Figure 1: Concept of Asymmetric Drift

Mathematical Drift Indicators

1. **Visual Drift (Δ_v):**

$$\Delta_v = \|\mathbb{E}_{x \in \mathcal{D}_t}[z_x] - \mathbb{E}_{x \in \mathcal{D}_{t-1}}[z_x]\|_2$$

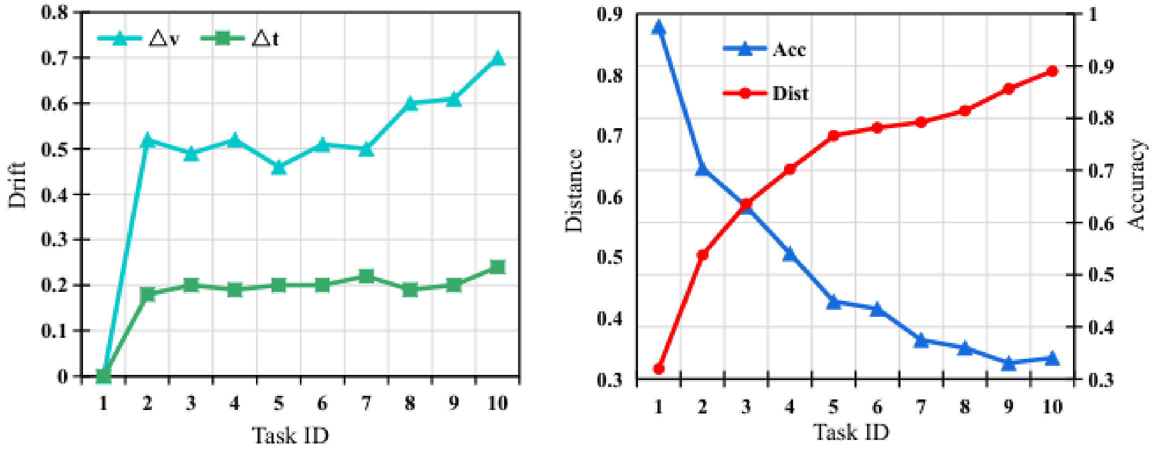
2. **Textual Drift (Δ_t):**

$$\Delta_t = \|\mathbb{E}_{y \in \mathcal{D}_t}[z_y] - \mathbb{E}_{y \in \mathcal{D}_{t-1}}[z_y]\|_2$$

3. **Cross-Modal Distance ($\mathrm{dist}(t)$):**

$$\mathrm{dist}(t) = \mathbb{E}_{(x,y) \in \mathcal{D}_t}\left[\left\| z_x^{(t)} - z_y^{(t)} \right\|_2\right]$$

1. Problem Definition: Empirical Proof of Asymmetric Drift



(a) Visual and textual drift.

(b) Distance vs. Accuracy.

Figure 3: Empirical evidence of Asymmetric Drift

Key Empirical Observations

- **Dominant Visual Shift ($\Delta_v \gg \Delta_t$):**
  Across all 10 tasks, Visual Drift (Δ_v) consistently and heavily exceeds Textual Drift (Δ_t).

- **Distance Expansion vs Accuracy Decay:**
  Cross-modal distance $\mathrm{dist}(t)$ increases monotonically over task updates, directly causing zero-shot accuracy to degrade over time.

1. Problem Definition: Why Existing Baselines Fall Short

Method Category	Core Mechanism	Primary Failure Cause
PEFT & Fine-Tuning *(EWC, LoRA-CL)*	Modifies visual backbone parameters per task	Unilateral visual updates aggravate Asymmetric Drift, corrupting joint representation space.
Prompt Tuning *(L2P, DualPrompt, CODA)*	Freezes backbone and learns prompt vectors	Prompts lack sufficient capacity to re-align cross-modal geometry under large domain shifts.
Replay / Pseudo-Gen *(SLCA, PROOF, ENGINE)*	Stores exemplars or generates pseudo-features	High computational overhead, privacy risks, and dependency on LLM prompt generation.

2. Theoretical Foundations: Canonical Correlation Analysis (CCA)

CCA identifies linear projection directions that maximize correlation between visual and textual feature spaces.

$$\max_{A,B} \operatorname{corr}(A^\top X, B^\top Y) = \frac{A^\top \Sigma_{XY} B}{\sqrt{A^\top \Sigma_{XX} A}\sqrt{B^\top \Sigma_{YY} B}}$$

Key Covariance Inputs

- Σ_{XX}, Σ_{YY}: Intra-modality covariance matrices of image features X and text features Y.
- Σ_{XY}: Inter-modality cross-covariance matrix between visual and text features.

Closed-Form Solution Benefit

- Derives a **shared correlated subspace** without modifying underlying backbone parameters.
- Re-aligns visual and textual distributions in closed form without gradient descent.

2. Theoretical Foundations: CCA Closed-Form Solution Steps

1. **Whitening Filter Calculation:**
   Perform Eigenvalue Decomposition on intra-covariances to obtain Whitening Filters $W_x = \Sigma_{XX}^{-1/2}$ and $W_y = \Sigma_{YY}^{-1/2}$.

2. **Whitened Cross-Covariance SVD:**
   Form whitened cross-covariance matrix $K = \Sigma_{XX}^{-1/2}\Sigma_{XY}\Sigma_{YY}^{-1/2}$ and solve SVD:

   $$K = V\Lambda U^\top$$

   where V, U are canonical directions and Λ contains canonical correlations.

3. **Canonical Subspace Projection Matrix Derivation:**
   Derive final linear projection matrices $A = \Sigma_{XX}^{-1/2}V$ and $B = \Sigma_{YY}^{-1/2}U$.
   Features mapped into aligned space ($z_x^{\text{cca}} = A^\top z_x, z_y^{\text{cca}} = B^\top z_y$) achieve maximal correlation.

3. Proposed Solution: Overall CCA-CL Architecture

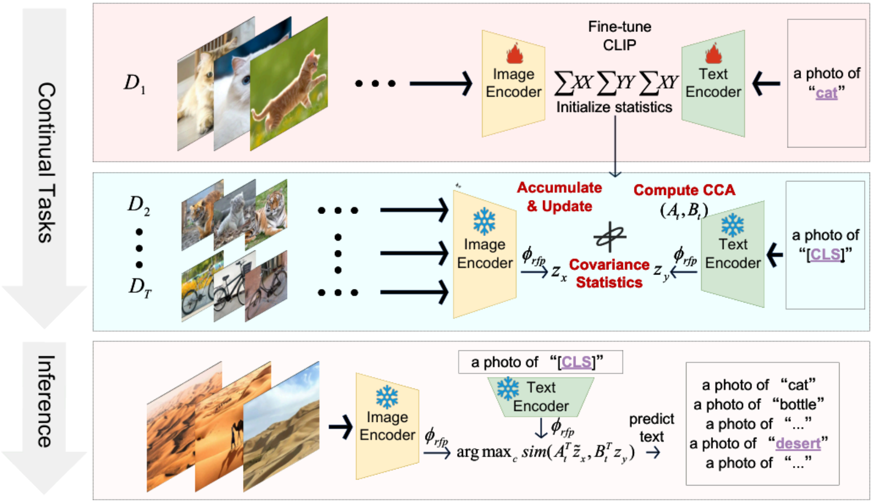


Figure 2: Overview of CCA-CL Framework

3-Stage Pipeline Overview

1. **First-Task Tuning (FTT):** Lightly tunes CLIP on Task 1 (5 epochs, lr=1×10^{-6}) for downstream domain adaptation.
2. **Frozen Parameter Backbone:** Freezes all CLIP parameters for subsequent tasks ($t \geq 2$) to eliminate further asymmetric drift.
3. **Closed-Form Subspace Alignment:** Accumulates visual-textual covariance statistics online and computes CCA projection matrices (A, B).

3. Proposed Solution: Exemplar-Free Covariance Accumulation

CCA-CL maintains statistical memory of historical data distributions without storing raw past images.

Traditional Replay Buffer

Stores raw past images ($x_1, x_2, \dots$) $\rightarrow$ violates privacy & adds memory overhead.

CCA-CL Statistical Memory

Accumulates 2nd-order covariances online $\rightarrow$ **100% Exemplar-Free** with zero sample storage.

Online Covariance Update Equations

For each incoming mini-batch of image features z_x and text features z_y:

$$\Sigma_{xx} \leftarrow \Sigma_{xx} + (z_x - \bar{z}_x)^\top (z_x - \bar{z}_x), \quad \Sigma_{xy} \leftarrow \Sigma_{xy} + (z_x - \bar{z}_x)^\top (z_y - \bar{z}_y)$$

3. Proposed Solution: Whitening & Energy Criterion Thresholding

To eliminate noise dimensions and compress subspace size, CCA-CL applies cumulative energy thresholding.

1. Whitened Decomposition

Decomposes matrix $K = CC^\top$ where $C = W_x \Sigma_{xy} W_y$:

$$KV = V\Lambda_\rho, \quad \text{where } \Lambda_\rho = \mathrm{diag}(\rho_1, \ldots, \rho_D)$$

2. Cumulative Energy Thresholding ($\eta = 0.99$)

Selects top r canonical directions satisfying:

$$\frac{\sum_{i=1}^{r} \rho_i^2}{\sum_{i=1}^{D} \rho_i^2} \geq \eta \quad (\eta = 0.99)$$

- **Dimensionality Reduction Effect:** Compresses feature subspace from $D = 6144$ down to average $\bar{r} \approx 47.2$, removing noisy dimensions while retaining 99% canonical energy.

3. Proposed Solution: Non-Linear RFP Extension

To overcome the linear limitations of classical CCA, Random Fourier Projection (RFP) maps features into a high-dimensional kernel space prior to alignment.

Non-Linear Kernel Mapping Formula

$$\phi_{\text{rfp}}(z) = \sqrt{\frac{2}{W}} \cos(W_{\text{rfp}} z + b)$$

- $W_{\text{rfp}} \sim \mathcal{N}(0, \sigma^2)$ (Random projection matrix)
- $b \sim \text{Unif}(0, 2\pi)$ (Random phase offset)

Core Advantages

- **Zero Trainable Parameters:** W_{rfp} and b are randomly fixed and never updated.
- **Kernel Approximation:** Approximates Gaussian RBF kernel without explicit kernel matrices.
- **Enhanced Capacity:** Expands projection width ($W = 6144$), enabling CCA to align complex non-linear structures.

3. Proposed Solution: Inference & Zero-Shot Classification

During inference, input features pass through frozen CLIP encoders, map via RFP, and project into the aligned CCA subspace.

$$\hat{y} = \arg\max_c \frac{\left(A^\top \phi_{\text{rfp}}(F_v(x))\right)^\top \left(B^\top \phi_{\text{rfp}}(z_y^{(c)})\right)}{\|A^\top \phi_{\text{rfp}}(F_v(x))\|_2 \left\|B^\top \phi_{\text{rfp}}(z_y^{(c)})\right\|_2}$$

Step-by-Step Inference Flow

1. **Extract Raw Features:** Image feature $z_x = F_v(x)$, Text prototype $z_y^{(c)} = F_t(\text{"a photo of [CLS]"})$.
2. **Apply Non-Linear RFP:** Compute $\phi_{\text{rfp}}(z_x)$ and $\phi_{\text{rfp}}(z_y^{(c)})$.
3. **Project & Predict:** Apply precomputed projection matrices A, B and evaluate cosine similarity across all candidate classes.

4. Experimental Results: SOTA Benchmark Comparison

Comparison across 4 major benchmarks under 10-task Class-Incremental setting (Backbone: ViT-B/16).

Method	Exemplar	CIFAR-100 (Avg / Last)	ImageNet-R (Avg / Last)	ImageNet-100 (Avg / Last)	CUB-200 (Avg / Last)
L2P (CVPR'22)	×	81.9 / 73.0	81.6 / 75.9	80.5 / 67.2	70.8 / 57.9
DualPrompt (ECCV'22)	×	81.4 / 72.5	82.0 / 75.7	80.6 / 67.3	69.8 / 57.4
Continual-CLIP (arXiv'22)	×	75.2 / 66.7	79.1 / 72.0	85.0 / 75.4	76.3 / 71.3
CODAPrompt (CVPR'23)	×	76.9 / 62.2	78.0 / 67.5	64.1 / 34.7	73.1 / 62.9
SLCA (ICCV'23)	✓	80.5 / 67.5	75.9 / 70.3	78.6 / 59.9	79.1 / 68.2
MagMax (ECCV'24)	×	85.6 / 79.0	87.1 / 80.8	86.3 / 75.9	70.8 / 62.1
RAPF (ECCV'24)	✓	86.1 / 79.0	85.5 / 80.2	87.5 / 80.2	83.0 / 76.3
PROOF (TPAMI'25)	✓	84.8 / 76.2	82.8 / 77.0	84.7 / 72.4	83.9 / 79.3
CCA-CL (no FTT)	×	**85.4 / 81.0**	**84.7 / 79.4**	**86.0 / 78.0**	**85.9 / 79.2**
CCA-CL (Ours)	×	**87.2 / 83.0**	**86.8 / 81.3**	**86.9 / 80.2**	**86.3 / 79.7**

4. Experimental Results: Modality Distance & Accuracy Trends

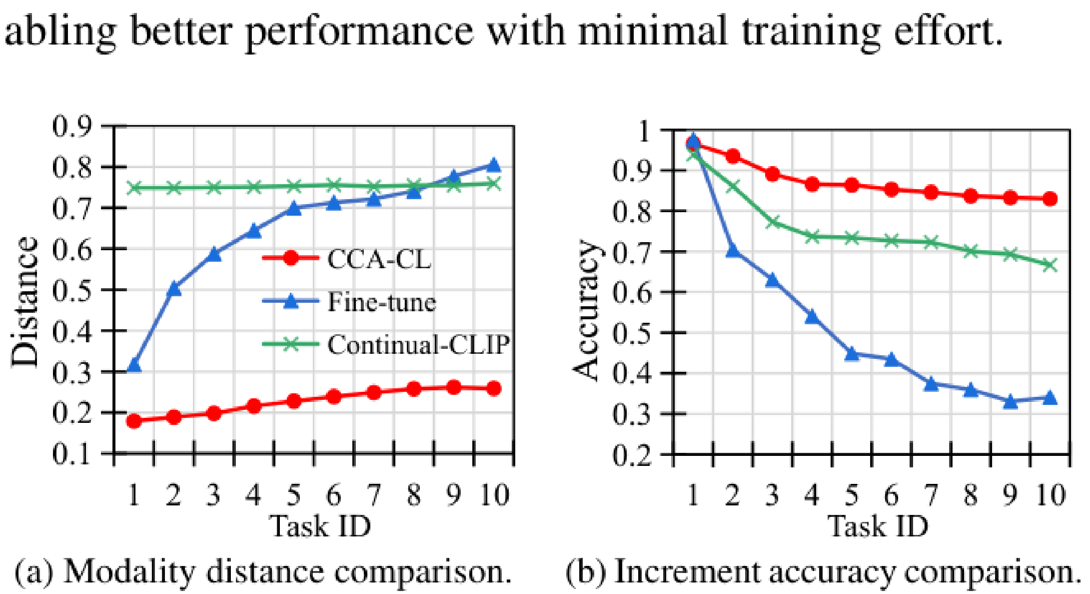


Figure 5: Modality Distance (a) and Increment Accuracy (b) across tasks

Baseline Behavioral Analysis

- **Fine-tune:** Modality distance expands continuously due to asymmetric drift, causing accuracy to collapse over tasks.
- **Continual-CLIP (Frozen):** Freezes all parameters; distance remains static but large, yielding sub-optimal accuracy (66.7%).
- **CCA-CL (Ours):** Subspace projection minimizes distance and prevents drift, consistently achieving highest accuracy (83.0%).

4. Experimental Results: Training Time & Computational Efficiency

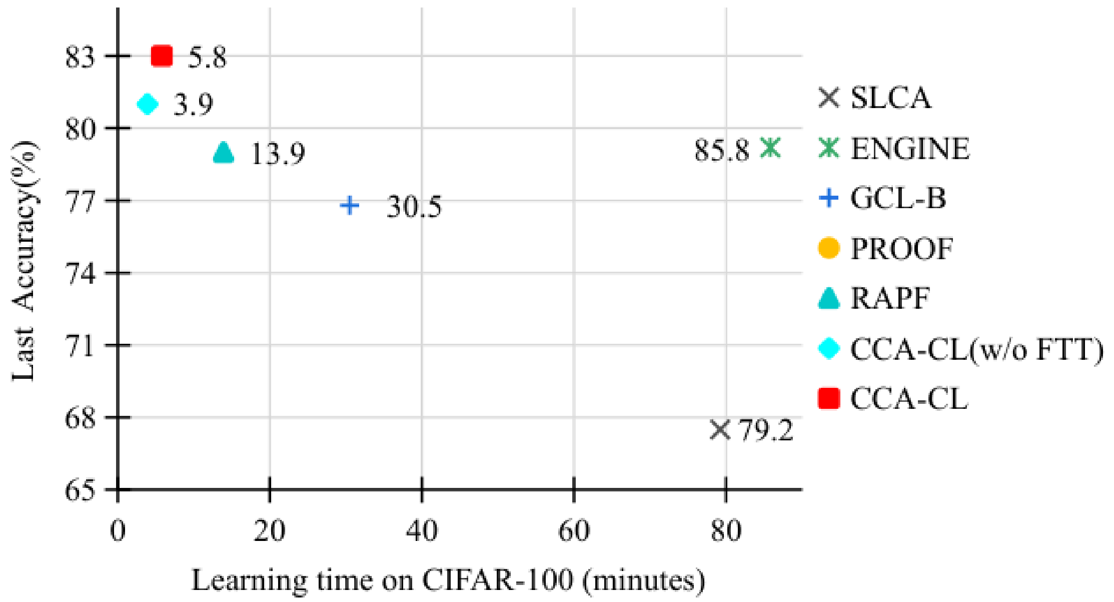


Figure 4: Learning Time vs Accuracy Trade-off on CIFAR-100

Runtime Efficiency Comparison

- **CCA-CL (Full):** Total training time of **5.8 minutes** (83.0% Accuracy).
- **CCA-CL (w/o FTT):** Total training time of **3.9 minutes** (81.0% Accuracy).
- **vs. SOTA Competitors:**
  - RAPF: 13.9 min (2.4x slower)
  - ENGINE / SLCA: >80 min (14x~20x slower)

4. Experimental Results: Ablation Studies

Ablation study on CIFAR-100 evaluating component contributions.

CCA	RFP	FTT	CIFAR-100 Last Acc (%)	Gain
×	×	×	66.7%	Base (Continual-CLIP)
✓	×	×	76.5%	**+9.8%** (Linear CCA)
✓	✓	×	81.0%	**+4.5%** (Non-linear RFP)
✓	✓	✓	**83.0%**	**+2.0%** (First-Task Tuning)

Component Impact Analysis

1. **CCA Alignment (+9.8%):** Closed-form subspace projection directly bridges the modality gap.
2. **RFP Extension (+4.5%):** Kernel mapping captures non-linear cross-modal dependencies.
3. **First-Task Tuning (+2.0%):** Adapts pre-trained representations to downstream data.

4. Experimental Results: Parameter Sensitivity Analysis

1. RFP Output Width W

W	Memory (GB)	Last Acc (%)
2048	3.1	79.7%
4096	4.3	80.3%
6144	**6.3**	**81.0%**
8192	8.8	80.6%
10240	11.9	80.2%

- **Optimal Width:** $W = 6144$ balances accuracy and memory footprint.

2. Energy Threshold η

η	Retained Dims ($\bar{r}$)	Last Acc (%)
0.90	34.0	76.5%
0.95	42.5	80.3%
0.99	**47.2**	**81.0%**
0.999	48.5	80.9%
1.00	6144.0	80.8%

- **Optimal Threshold:** $\eta = 0.99$ retains top $\bar{r} = 47.2$ dimensions, filtering noise.

Conclusion & Key Takeaways

1. **New Problem Formulation:**
   Identified and mathematically defined **Asymmetric Drift (AD)** as a fundamental bottleneck in unimodal adaptation of Vision-Language Models.

2. **Statistically Grounded Solution:**
   Proposed **CCA-CL**, combining closed-form Canonical Correlation Analysis with Random Fourier Projection (RFP) for non-linear subspace alignment.

3. **State-of-the-Art Accuracy & Efficiency:**
   Achieved SOTA accuracy across 4 benchmarks while remaining **100% Exemplar-Free** and running **up to 20x faster** than existing baselines.

Thank You! Q & A